Assignment No.1

Q1. What is a variable in C/C++ ?
Ans. A variable is a named memory location used to store data.

Q2.Why we use variable in programs?
Ans. We use variable in program because store and manuplate the data in program.

Q3. What is the difference between a variable name and its value?
Ans. A variable name is the identifier, while its value is the data sotred in it.

Q4. Which of the following is a valid variable name?
a)2num b)total_marks c)for d)my marks
Ans. b)total_marks

Q5.Why is Age different form age?
Ans.Because c and c++ are case sensitive ,Age and age are treated as different variable.

Q6.Can a variable name contain spaces? Why?
Ans. No, because spaces separate different tokens.

Q7. Can variable name start with an underscore(_)?
Ans.yes, variable name start with an underscore(_).

Q8.what happens if two variables have the same name in the same scope?
Ans.it causes a compilation error.

Q9.Write three meaningful variable names for storing student information?
Ans.rollno, studentName,class.

Q10.which naming convention is better and why: studentMarks or sm?
Ans.studentMarks naming convention is better because it is more meaning full and easy to understand.

Q11.what is a data type?
Ans.A data type define which type of data a variable can store.

Q12.which data type is used to store whole numbers?
Ans. The data type name is int.

Q13. data type is used to store decimal value?
Ans.float or double is to store decimal value.

Q14.what is the difference b/w float and double?
Ans. double size is greater than float size after decimal.

Q15. Which data type store a single character?
Ans. chat is used to store a single character.

Q16. Which data type stores True/false values?
Ans. bool is used to stores True/false values.

Q17.What data type is suitable for storing a person’s name?
Ans. String is suitable for storing a person’s name.

Q18.Predict the output: int age=18;cout<< age;?
Ans. the output will be 18.

Q19. Identify the data type:25,3.14,’A’,true?
Ans . 25 int,3.14 float, ‘A’ char, true bool.

Q20.What happens if you store a decimal number in an int variable?
Ans. in which only whole number is stored and decimal part is discarded.

Q21. What is the purpose of cin?
Ans. cin is used to take input from the user.

Q22. What is the purpose of cout?
Ans. cout is used to take output from the user.

Q23.which symbol is used with cin?
Ans.the >> symbol is used with cin.

Q24.which symbol is used with cout?
Ans.the << symbol is used with cout.

Q25. Explain cin>>age;
Ans . it takes a value entered by the user and store it in the variable age.

Q26.Explain : cout<<age;
Ans. it display the value of the variable age on the screen.

Q27. What is the purpose of endl?
Ans. endl move the cursor to the next line.

Q28. Write a statement to input a student’s marks?
Ans. cin>>marks;
Q29. Write a statement to display “Welcome to c++”?
Ans. cout<<”Welcome to c++”;

Q30. Predict the output: cout<<”Programming”;
Ans. the output will be Programming.

Q31.What is an operator?
Ans. An operator is a symbol used to perform operations on data.

Q32. Which operator is used to Addition?
Ans. The + operator is used to Addition.

Q33. What is the result of 10%3?
Ans.   The result will be 1. this (%) operator gives remainder.

Q34. What is the result of 8%2?
Ans.   The result will be 0. this (%) operator gives remainder.

Q35. Which Operator checks equality?
Ans. The “==” operator checks equality.

Q36. What is the difference b/w = and ==?
Ans. = is used for assignment, while == is used for comparision.

Q37. What is the result of 5>3?
Ans. true.

Q38. What does && represent?
Ans. && represents the Logical And operator.

Q39. What does || represent?
Ans. the || represents the Logical OR operator

Q40. What does ! represent?
Ans. ! represents the Logical NOT operator.

Q41.What a program to print “Hello World”?
Ans. 1.Problem Statement:
write a program to print”Hello World”.
2.Algorithe/steps:
Start.
Display “Hello World”.
Stop.
3.Dry Run with sample input:
No input required.
4. Program code(c++):

```
#include<iostream>
using namespace std;
int main(){
cout<<"Hello World";
return 0;
}
```

Output-
Hello World

== Code Execution Successful ==

6. Explanation of the Logic Used:
The program uses cout to display" Hello World" on the screen.

Q42. Write a program to input two numbers and display their sum?
Ans. 1.Problem Statement:
Write a program to input two numbers and display their sum.
2.Algorithe/steps:
Start.
Input two numbers.
Calculate their sum.
Display the sum.
Stop.
3.Dry Run with sample input:
Input: 34,23;
Sum=57
4. Program code(c++):

```
#include<iostream>
using namespace std;
int main(){
int a,b;
cout<<"Enter two numbers :";
cin>>a>>b;
cout<<"sum="<<a+b;
return 0;
}
```

Output-
Enter two numbers : 34
23
Sum=57
== Code Execution Successful ==

6. Explanation of the Logic Used:
The program takes two numbers as input, adds them, and displays the result.

Q43. Write a program to calculate the area of a rectangle?
Ans.
1.Problem Statement:
Write a program to calculate the area of a rectangle.
2.Algorithe/steps:
Start.
Input length and breadth.
Calculate Area=Length*Breadth;
Display the area.
Stop.
3.Dry Run with sample input:
Length=6,Breadth=3;
Area=18;

4. Program code(c++):

```
#include<iostream>
using namespace std;
int main(){
int Length,Breadth;
cout<<"Enter length and breadth";
cin>>Length>>Breadth;
cout<<"Area="<<Length*Breadth<<endl;
return 0;
}
```

Output-
Enter Length and Breadth 6
3
Area=18

== Code Execution Successful ==

6. Explanation of the Logic Used:
The program multiplies the length and breadth to find the area of the rectangle.

Q44. What formula is used to calculate Simple Interest?
Ans. SI=(P*R*T)/100
Where:
SI=simple interest

P=Principal Amount (the original amount of money)
R= Rate of interest (percentage per year)
T=Time.

Q45. Write a program to calculate Simple Interest?
Ans.
1.Problem Statement:
calculate Simple Interest.
2.Algorithe/steps:
Start.
Input = Principal, Rate and time
Calculate SI = (P*T*R)/100.
Display=SI
Stop.
3.Dry Run with sample input:
P=2000, R=3, T=3
SI=(2000*3*3)/100
SI=180
4. Program code(c++):

```
#include<iostream>
using namespace std;
int main(){
float P,R,T,SI;
cout<<"Enter value of  P,R,T =";
cin>>P>>R>>T;
SI = (P*T*R)/100
Cout<<"Simple Interest="<<SI<<endl;
return 0;
}
```

Output-
Enter value of P,R,T =2000 3 3
Simple Interest=180

== Code Execution Successful ==

Q46.Why is a temporary Variable used while swapping two numbers?
Ans. A temporary variable is used to store one value temporary so that it is not lost during swapping.

Q47. Swap the values: a=10,b=20.
Ans.

. 1.Problem Statement:
Write a program to swap the values of two variables using a temporary variable.
2.Algorithe/steps:

1. Start
2. Declare variables a,b,and temp.
3. Assign a=10 and b=20.
4. Store the value of a in temp.
5. Assign the value of b to a .
6. Assign the value of temp to b.
7. Display the swapped values.
8. Stop

3.Dry Run with sample input:

Steps	a	b	temp
Initial	10	20	-
Temp=a	10	20	10
a=b	20	20	10
b=temp	20	10	10

4. Program code(c++):

```
#include<iostream>
using namespace std;
int main(){
int  a=10,b=20,temp;
temp=a;
a=b;
b=temp;
cout<<"After swapping:"<<endl;
cout<<"a="<<a<<endl;
cout<<"b="<<b<<endl;
return 0;
}
```

Output-
After Swapping :
a=20
b=10

== Code Execution Successful ==

6. Explanation of the Logic Used:

The program uses a temporary variable temp to store the value of a so that it is a not lost while exchanging the values of a and b.

Q48. What formula is used to convert Celsius into Fahrenheit?
Ans. Fahrenheit= (Celsius*(9/5))+32

Q49. What is the Fahrenheit value of 0°C ?
Ans. the Fahrenheit value of 0°C is 32°F.

Q50. Why can integer division give incorrect results in the Celsius-to-Fahrenheit program?
Ans. Integer division discards the decimal part, which can lead to inaccurate Fahrenheit values.